

**V.V.P. INSTITUTE OF ENGINEERING &
TECHNOLOGY, SOLAPUR**



A MINI PROJECT REPORT ON

AGROVISION.AI: INTELLIGENT

AGRICULTURAL ANALYSIS

PLATFORM

SUBMITTED BY:

Student Name: _____

PRN: _____

UNDER THE GUIDANCE OF:

Guide Name: _____

ACADEMIC YEAR 2025-2026



CERTIFICATE

This is to certify that the mini project entitled "**Agrovision.Ai: Intelligent Agricultural Analysis Platform**" has been successfully completed by _____
(PRN: _____) in partial fulfillment of the requirements for the diploma.

Project Guide

Head of Department

ABSTRACT

In the agricultural sector, rapid and accurate diagnosis of plant diseases and soil conditions is critical for maximizing yield. This project, **Agrovision.Ai**, presents an intelligent agricultural analysis platform designed to serve as a virtual agronomist. The core innovation of this system relies on a **Custom Trained Machine Learning Model (Convolutional Neural Network)** that was personally trained on a diverse dataset of local crop images to specifically identify regional plant diseases with high accuracy.

Users interact with the system by uploading a photo of their farm or crop and sharing their GPS coordinates. The system processes the image using our custom CNN model for precise disease detection, whilst fetching real-time environmental context such as localized weather data (via Open-Meteo) and soil chemical properties (via SoilGrids). All these data points are synergized and passed through Google Gemini 2.5 Vision AI to formulate actionable, holistic crop recommendations.

This hybrid AI approach ensures extreme accuracy by combining specialized local disease modeling with general-purpose generative formulation. The final output is a downloadable PDF report providing actionable guidance on fertilizer use, irrigation scheduling, and disease treatment, bringing precision agriculture to smallholder farmers seamlessly.

CHAPTER 1: INTRODUCTION

1.1 Background

Agriculture remains the backbone of the economy, yet modern farming tools are often restricted to large-scale enterprises due to high costs. Agrovision.Ai was developed to bridge this gap by bringing advanced diagnostics to the mobile phones of everyday farmers.

1.2 Custom Trained AI Model Integration

Unlike standard applications that rely entirely on generic API endpoints for AI, Agrovision.Ai distinguishes itself through the implementation of a **Personally Trained Machine Learning Model**. We trained a Convolutional Neural Network (CNN) specifically tailored to the agricultural anomalies of this region. This model pre-processes the uploaded crop image, ensuring that local diseases are accurately identified before the data is handed over to the secondary Google Gemini formulation layer. This guarantees a localized, highly accurate response.

CHAPTER 2: SYSTEM ARCHITECTURE

2.1 Core Workflow

- **Input:** The farmer uploads a crop image (encoded in Base64) and provides GPS coordinates.
- **Parallel Data Fetching:** The backend API utilizes `Promise.all()` to simultaneously fetch Open-Meteo weather data and SoilGrids chemistry data, significantly reducing latency.
- **Custom CNN Processing:** The image is analyzed by our internally trained ML model to detect diseases, pests, and soil textures.
- **Gemini Synergy:** The diagnosis from the custom model is combined with the environmental data and analyzed by Gemini 2.5 Vision for final treatment generation.
- **Automated PDF Reporting:** The final actionable insights are dynamically rendered into a PDF using `pdf-lib` and emailed to the user instantly.

2.2 Technology Stack

The platform is built on **Next.js 14** for a fast, serverless architecture. **Supabase (PostgreSQL)** is employed for secure data storage and Row-Level Security, ensuring all farmer reports remain private.